



Laminar burning velocities of methane + formic acid + air flames: Experimental and modeling study

Marco Lubrano Lavadera, Alexander A. Konnov*

Division of Combustion Physics, Lund University, SE-22100 Lund, Sweden



ARTICLE INFO

Article history:

Received 16 September 2020

Revised 27 October 2020

Accepted 28 October 2020

Keywords:

Formic acid

Methane

Burning velocity

Kinetic model

ABSTRACT

Laminar burning velocities of methane + formic acid + air flames were determined using the heat flux method at 1 atm and initial gas temperature of 353 K. For fuel mixtures containing 75 or 50% mole fraction of HCOOH, a range of equivalence ratios from 0.7 to 1.3 was covered. In stoichiometric mixtures burning velocities were measured varying fuel composition from pure methane to 85% of formic acid. Due to the relatively high initial temperature close to the operational limit of the present burner, an extrapolation procedure had to be used leading to rather high experimental uncertainties up to ± 2 cm/s. New measurements have been compared with the predictions of the recent kinetic model of the authors and the model of Glarborg et al. [34]. Both models failed, and the reasons for the disagreement were analysed. A modification of the rate constant of reaction $\text{HOCO}(\text{+M})=\text{H}+\text{CO}_2(\text{+M})$ was suggested that dramatically improved the present model performance. This model was then compared with the predictions of the model developed by Marshall and Glarborg [19] at the conditions of the available experimental data for HCOOH flames. The comparison indicates some inconsistencies in the recent measurements of the burning velocity of formic acid.

© 2020 The Author(s). Published by Elsevier Inc. on behalf of The Combustion Institute.
This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>)

1. Introduction

Carboxylic acids, in particular formic (HCOOH) and acetic (CH_3COOH) acids, were recognized as ubiquitous pollutants detected in the troposphere [1], rural continental [2], and urban atmosphere [3,4], rain water [5], etc. The anthropogenic source of carboxylic acids was linked to exhaust of diesel [6] and spark-ignition engines fuelled by commercial gasoline [3], hydrocarbon mixtures [7], and natural gas [8]. Further laboratory studies [9,10] confirmed the formation of carboxylic acids in premixed flames of hydrocarbons that motivated the development of the first detailed kinetic model by Battin-Leclerc et al. [11] aimed at the prediction of the acids' formation in flames. Although an encouraging agreement was found with the scarce experiments for propane flames [9], validation and improvement of the model were hampered by the lack of experimental studies on the combustion of carboxylic acids.

A transition to sustainable energy production and renewable fuels revitalises interest in the flame chemistry of carboxylic acids. First, in flames of methanol, which is a commercially available fuel supplement, formic acid was identified as the abundantly pro-

duced intermediate [12]. Second, bio-oils envisioned as alternative fuels and produced from biomass (e.g., wood) pyrolysis contain significant fractions of formic and acetic acids [13,14]. Furthermore, formic acid was suggested as a hydrogen energy carrier [15–18] which could be used for storage, delivery, and on-site production of hydrogen. These aspects inspired recent experimental and modeling studies on acids' flames.

Marshall and Glarborg [19] developed a detailed kinetic model for formic acid oxidation combining available literature rate constants and also calculating several rate constants of the acid with radicals. The only target for this model validation was rather old measurements of the burning velocity by De Wilde and Van Tiggelen [20]. For acetic acid, the flame structure of premixed low-pressure flames was investigated [21], and laminar burning velocities were obtained using the heat flux method [22]. Also, burning velocities of acetic and propionic acids have been measured in spherical flames [23]. Most recently, several groups performed dedicated studies of the flames of formic acid at atmospheric pressure summarized in Table 1. Sarathy et al. [24] implemented two similar experimental setups for measurements in spherical flames, however, employed different (linear and non-linear) stretch-correction approaches for data processing. Together with the investigation of formic acid + air flames at 373 and 423 K, the fuel mixtures of HCOOH + (H_2 + CO_2) were also studied.

* Corresponding author.

E-mail address: alexander.konnov@forbrf.lth.se (A.A. Konnov).

Table 1
Laminar burning velocity measurements of formic acid flames.

Fuel	Oxidizer	Oxygen content	Temperature, K	Method	Source
HCOOH	O ₂ +N ₂	76–88%	433	Bunsen burner	[20]
HCOOH	air	21%	373, 423	Spherical flames	[24]
HCOOH+H ₂ +CO ₂	air	21%	373, 358	Spherical flames	[24]
HCOOH	O ₂ +N ₂	30, 35, 40%	423, 453	Spherical flames	[25]
HCOOH	O ₂ +Ar	variable	368	Nozzle burner	[26]
HCOOH+CH ₄	air	21%	353	Heat flux	p.w.

These mixtures mimic pre-decomposition of formic acid that, as expected, improves the reactivity of the fuel and increases its burning velocity. Two detailed kinetic models were tested: the original model of Marshall and Glarborg [19] (in the following MG model) and a model merged by Sarathy et al. [24] from the AramcoMech 2.0 and a formic acid reaction subset from [19]. This compiled model was found in somewhat better agreement with the new burning velocities [24] compared to the MG model, yet the measurements at 423 K and 373 K were under- and overpredicted, respectively, especially in rich mixtures.

Yin et al. [25] measured burning velocities of formic acid in artificial air containing from 30 to 40% of oxygen at 423 and 453 K using the spherical flame method with non-linear stretch correction. They found that the MG model matches well their experimental results. AramcoMech 3.0 contains radicals HOCO, and HCOO (HC(O)O), though they are not formed from formic acid; nevertheless, Yin et al. [25] performed an extensive comparison of these two kinetic schemes at the conditions of their experiments. Replacing different parts of the AramcoMech 3.0 (transport properties, thermodynamic data, and pertinent reactions) by corresponding parameters from the MG model, it was demonstrated that complete merging yields the mechanism which notably underpredicts the burning velocities of formic acid. This observation is in line with the behaviour of the merged model (AramcoMech 2.0 + MG model) suggested by Sarathy et al. [24].

Osipova et al. [26] focused their study on the investigation of the flame structure of premixed HCOOH+H₂+O₂+Ar flames at atmospheric pressure using molecular beam mass-spectrometer. They compared concentration profiles of the reactants, intermediate species, and products with predictions of three kinetic schemes: the MG model, the early model developed for acetic acid flames by Christensen and Konnov [22], and the compiled model of Sarathy et al. [24]. The room for improvement of each of these models was indicated and discussed. In addition, the burning velocities of the HCOOH+O₂+Ar mixtures at 368 K were measured on a Mache-Hebra nozzle burner. In these measurements, the argon fraction was fixed at 0.55, while amounts of formic acid and oxygen varied according to the equivalence ratio. Two models [19,24] satisfactorily reproduced burning velocities of these flames deviating mostly in rich mixtures, while the acetic acid mechanism [22] failed, showing overprediction by a factor of 2.

Meanwhile, the earlier mechanism of Christensen and Konnov [22] was elaborated via theoretical studies mostly focused on reactions of ketene, which is the key intermediate in acetic acid flames [27–29]. Moreover, Savchenkova et al. [29] updated the formic acid subset largely following recommendations of Marshall and Glarborg [19]. To avoid confusion with other models developed in Lund, this mechanism [29] is in the following called SK model. The revision of the formic acid reactions was prompted by the fact that HCOOH appears as a minor intermediate, however, none of the updated reactions was found influencing the burning velocities of diacetyl or acetic acid flames and further validation was not performed. Therefore, the goals of the present study were: (a) to validate the SK model using available burning velocity measurements for formic acid flames; and (b) to produce new experimen-

tal data employing the heat flux method which requires no stretch correction. Fuel mixtures of formic acid + methane were investigated in the present work for two reasons: first, the limitations of our experimental installation described in the following do not allow measuring pure formic acid + air flames. Second, as it was discussed since early studies of formic acid flames [20], very low reactivity of formic acid requires special measures to realise stable combustion. This could be achieved, as exemplified in previous works, by increased concentration of O₂ in the oxidizer [20,25], replacing N₂ by Ar [26], or mixing formic acid with highly reactive fuels [24]. In a way similar to what has been proposed for ammonia utilization as a fuel supplement, one may suggest burning formic acid with natural gas, that was explored in the present study.

2. Experimental details

The burning velocity measurements have been performed at atmospheric pressure using the heat flux method [30]. This flat-flame-based technique offers important advantages over other methods since it creates stationary, premixed laminar flames that, at suitable conditions, can be considered stretchless, adiabatic, and one-dimensional. A review of the method and its development, detailed descriptions of the experimental setup, data processing algorithms, and uncertainty analysis can be found elsewhere [31], thus only the relevant elements are outlined below.

All heat flux burners employed so far in our laboratory are made of brass and it is well known that formic acid can be corrosive towards such a material. Hence, the burner with the worst performances due to an old design, designated as burner “E2” by Alekseev et al. [31] has been employed. This burner was successfully used by Christensen and Konnov [22] to determine the laminar burning velocities of acetic acid + air flames. In that study [22], once finalized the experiments with acetic acid, a green patina was found covering the surface of the burner head. However, no noticeable consumption of acetic acid or formation of unexpected products close to the burner surface was reported. It is important to note that in the present experiments the formation of the green oxidation layer was not observed.

The heat flux burner used in the present work consists of a 2 mm-thick plate attached to the burner head and perforated with 0.5 mm-diameter holes at a pitch of 0.7 mm. The burner plate has a nominal diameter of 3 cm, but the exact value of the effective perforation area is unknown. Therefore, to determine it, a procedure similar to that employed in [22] has been adopted, which however leads to a higher uncertainty of the measured burning velocity than commonly reported for measurements performed with the heat flux method (usually <1 cm/s). The burner plate is surrounded by a heating jacket supplied with thermostatic water to keep its edge at a constant temperature of 368 K, while the temperature of the fresh gas mixture has been maintained at 353 K (±1 K) by a separate heating system in the plenum chamber, also supplied with water. This temperature has been chosen in order to cover a wide range of equivalence ratios, which is constrained by the vaporization limit of formic acid.

The positive temperature difference between the burner plate and the inlet mixture allows for the stabilization of flat adiabatic flames since the amount of heat lost by the flame to the burner is compensated by the gain to the unburned mixture as it enters the preheated burner plate. The laminar burning velocity was determined by varying the flow rate of the gas mixture until a uniform radial temperature distribution over the burner plate, recorded by eight E-type shielded thermocouples (0.5 mm diameter) inserted into specific spaced holes over the burner plate, was achieved. For this condition, the net heat flux from/to the burner plate is zero and the velocity of the unburned gas is equal to the adiabatic laminar burning velocity. Hermanns [32] found that the temperature difference between the burner plate and the unburned gas mixture should be higher than 30 K to avoid flame instabilities. However, in the present experiments, this ΔT has been set to 15 K, as explained, due to the vaporization limit. At these temperatures, most of the investigated flames showed corrugation or cellularity around adiabatic conditions. Therefore, the burning velocity was determined by extrapolation from flat sub-adiabatic flames. This procedure leads to higher experimental errors compared to the conventional interpolation procedure. An example of how the burning velocity is determined from sub-adiabatic flames, as well as an evaluation of the associated uncertainties, can be found in [31].

A mixing panel provides controlled flows of fuel and air to set the required equivalence ratio. The liquid fuel feeding system consists of a stainless steel cylinder pressurized with nitrogen to feed the formic acid through a Coriolis liquid mass flow controller into a Controlled Evaporator Mixer (CEM), both made by Bronkhorst High-Tech. The total uncertainty of the liquid flow rate is a sum of 0.5 g/h plus the stated flow repeatability of the liquid flow controller, which corresponds to 0.2% of the reading. Formic acid was used as delivered from Sigma Aldrich in a sealed bottle (98–100% purity). Synthetic air was delivered by Linde (21% O_2 , 1% relative uncertainty) and its required flow rates were metered by means of a Bronkhorst High-Tech digital thermal mass flow controller. In order to prevent fuel condensation throughout the fueling system, an electrically heated tube was used to connect the evaporator to the plenum chamber and its temperature was set equal to the temperature of the plenum chamber (353 K). In our laboratory, two mixing panels with different CEM capacities are currently available. In the first one, the CEM has a fuel capacity of 200 g/h, which is too low to cover the flows if formic acid + air flames have to be investigated. Note that the flow rates of formic acid and air are fixed by the combination of stoichiometry and laminar burning velocity. Therefore, the second mixing panel with a larger fuel capacity (1200 g/h) was used. However, in this second setup, the air flow is constrained by a lower limit of 5 nL/min, which is much higher compared to the first installation (0.2 nL/min). After running the experiments, it turned out that this lower limit was too high to cover the flows of formic acid + air flames, given their unexpectedly low burning velocities. Due to these limitations, fuel mixtures of formic acid + methane were investigated in the present work, as mentioned in the introduction. The use of methane as a fuel component allows to increase the burning velocity and thus to increase the air flow rates to values higher than 5 nL/min. Similar to air, the required methane flow rates were metered by means of a Bronkhorst High-Tech digital thermal mass flow controller. Both gas flow meters (air and methane) were calibrated before the measurements using a MesaLabs Definer 220 positive displacement calibrator. The total uncertainty of the gas flow rates is a sum of 1% stated accuracy of the calibrator plus the stated flow repeatability of the mass flow controllers, which corresponds to 0.2% of the set operating conditions.

Detailed analysis and quantification of experimental uncertainties was reported earlier [31] and the overall accuracy of burn-

ing velocity in the present measurements was estimated to be between 0.8 and 2.2 cm/s, depending on the experimental conditions. The equivalence ratio of the mixtures was varied from 0.7 to 1.3 with a maximum uncertainty of 0.02 and the mole fraction of formic acid in the fuel mixture was varied from 0 (pure methane) to 0.85 with a maximum uncertainty of 0.0028.

3. Modeling details

Premixed flame modeling as well as sensitivity analysis were performed using ANSYS Chemkin 17.0 software [33]. The laminar burning velocity was calculated considering multicomponent transport and thermal diffusion options. A grid-independent solution was ensured by setting the parameters GRAD and CURV to 0.03, resulting in a number of grid points around 500–700 over the domain of 3 cm.

In addition to the MG model and the SK model mentioned above, a widely used recent mechanism of Glarborg et al. [34] (in the following GK model) was tested in the present work since the original formic acid model [19] does not include methane and heavier species needed for $HCOOH + CH_4$ flame modeling. The SK model was lately updated with reactions of propanol isomers [35] and NO_x sub-mechanism [36], which was further refined by Han et al. [37]. The present mechanism after modifications described in the following is available in the Supplemental Material.

4. Results and discussion

Laminar burning velocities of fuel mixtures (0.75 $HCOOH + 0.25 CH_4$) and (0.5 $HCOOH + 0.5 CH_4$) burning with air at 353 K are presented in Figs. 1 and 2, respectively. In both cases, the maximum burning velocity is found in slightly rich flames close to equivalence ratio 1.05. Also shown in these figures are predictions of three kinetic mechanisms: SK model, GK model, and the present model (including the case with the rate constant of HOCO decomposition from Golden et al. [38]), which will be discussed

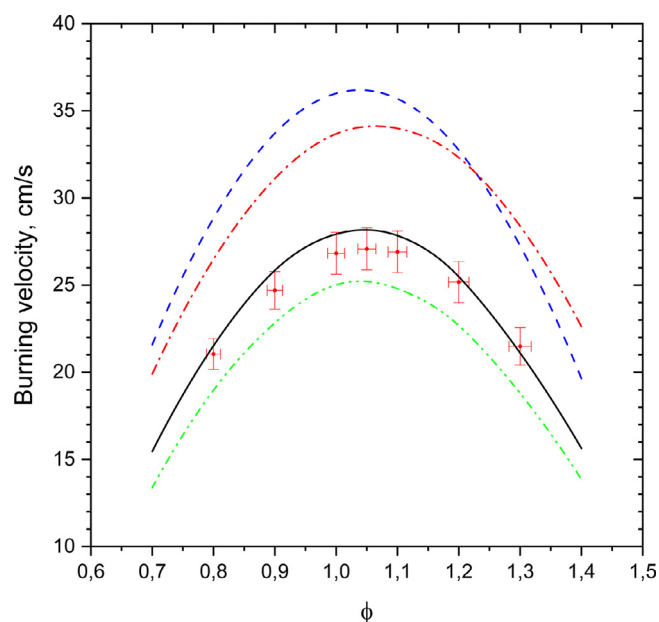


Fig. 1. Laminar burning velocities of (0.75 $HCOOH + 0.25 CH_4$) + air flames at 353 K. Symbols: present experiments; dashed blue line – SK model, dash-dot red line – GK model, green dash-double dot line – present model with the rate constant of HOCO decomposition from Golden et al. [38], solid black line – present model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

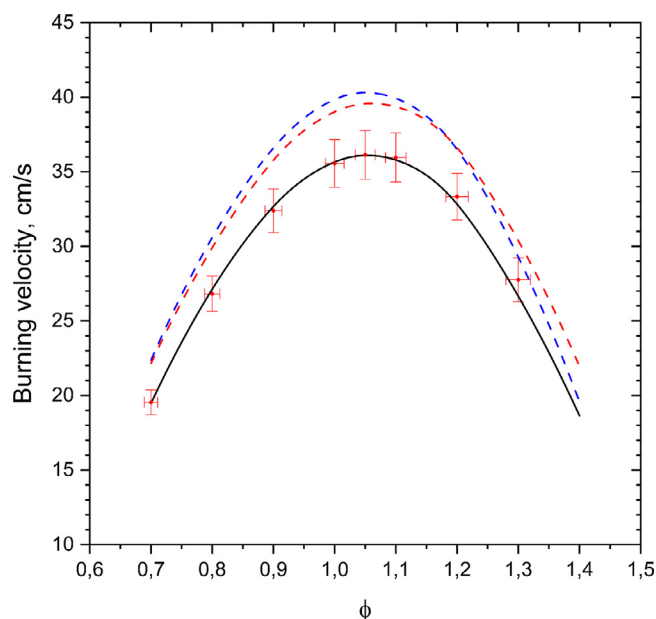


Fig. 2. Laminar burning velocities of (0.5 HCOOH + 0.5 CH₄) + air flames at 353 K. Symbols: present experiments; dashed blue line – SK model, dash-dot red line – GK model, solid black line – present model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

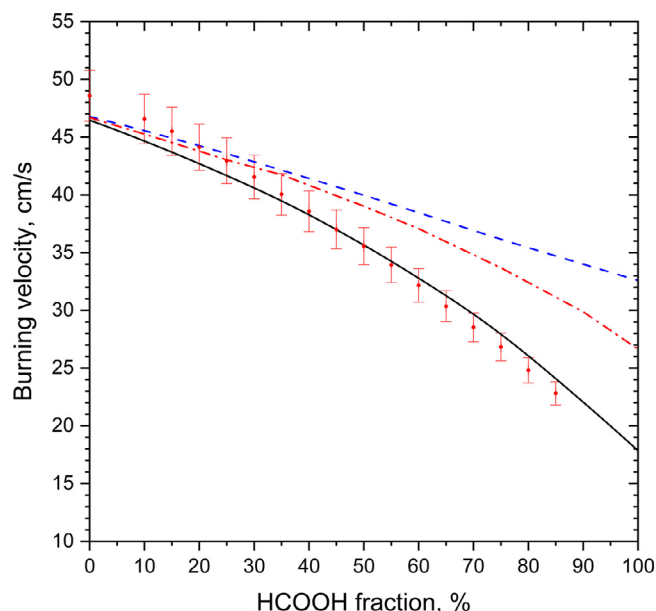


Fig. 3. Laminar burning velocities of stoichiometric HCOOH + CH₄ + air flames at 353 K. Symbols: present experiments; dashed blue line – SK model, dash-dot red line – GK model, solid black line – present model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

later. Both the SK model and the GK model significantly overpredict experimental data, which for the latter mechanism was an unexpected finding. The behaviour of the GK model was highly surprising since it accurately reproduces the burning velocities of methane flames as recently confirmed by Han et al. [37]. On the other hand, the original formic acid model from Marshall and Glarborg [19] was, at a first glance, incorporated in the GK model. This deviation of the GK model from the present experimental data is even better seen in Fig. 3, which shows laminar burning velocities of stoichiometric HCOOH + CH₄ + air flames at 353 K as a

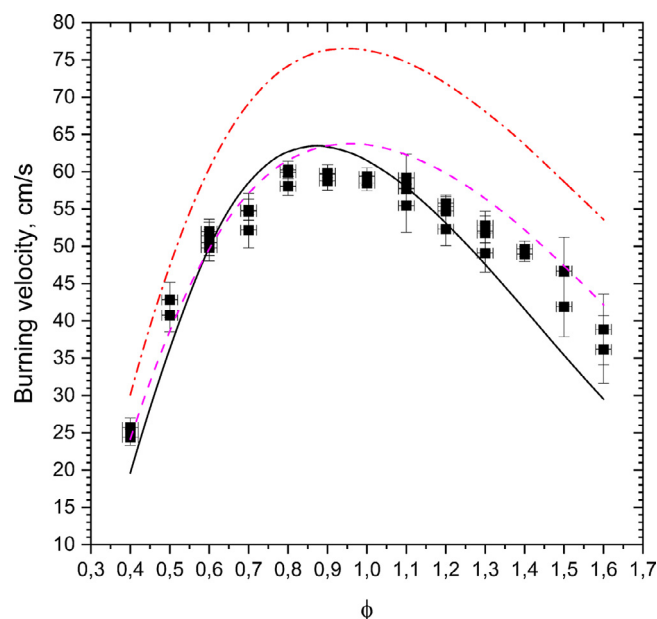


Fig. 4. Laminar burning velocities of HCOOH + (0.4 O₂ + 0.6 N₂) flames at 453 K. Symbols: experiments [25]; dash-dot red line – GK model, dashed magenta line – MG model, solid black line – present model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

function of the mole fraction of formic acid in the fuel mixture. Increasing replacement of methane by formic acid significantly reduces the burning velocity, while the SK model and GK model fail to reproduce this trend correctly.

To elucidate the reasons for the discrepancy between the MG model and the GK model, the burning velocities of HCOOH + O₂ + N₂ mixtures with 40% of O₂ in the oxidizer have been modeled and compared with the experimental data of Yin et al. [25] as shown in Fig. 4. The MG model is in a good agreement with these measurements, as was observed earlier [25], while the GK model significantly overpredicts them. One-by-one comparison of the reactions and rate constants implemented in the MG model and GK model revealed several differences outlined below. First, the rate constant of reaction



presented as a sum of two Arrhenius expressions was refitted in the GK model to avoid using negative pre-exponential factor implemented in the MG model. The total rate constant of this reaction, however, remained the same and thus does not affect calculated burning velocity. Second, the rate constant of reaction



was indeed modified as described in the Supplemental Material. Another change introduced in the GK model was the removal of HOCHOH formation reaction



and of couple reactions of its consumption via decomposition and oxidation by O₂. A direct comparison of the predictions of the original MG model with and without reactions of HOCHOH or with different rate constant of reaction (2) shows a very small impact (less than 0.1 cm/s) on the calculated burning velocities of formic acid flames.

For several species important in formic acid flames their thermodynamic data were also updated in the GK model. These

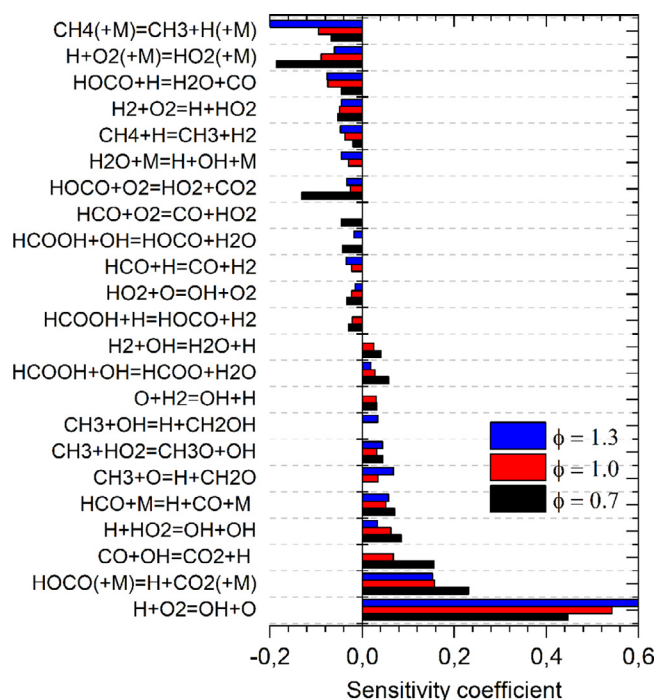


Fig. 5. Normalized sensitivity coefficients for the burning velocity of (0.75 HCOOH + 0.25 CH₄) + air flames at 353 K.

changes have a dramatic effect on the calculated burning velocity mostly due to the modifications of the thermodynamic data of HOCO radicals and formic acid. It should be noted that there is a discontinuity in the temperature dependence of enthalpy of HCOOH when thermodynamic data from the Burcat database updated in 2014 [39] are used (e.g., in the GK model) as illustrated in Fig. S7 in the Supplemental Material. The earlier data from the same database of 1988 implemented in the MG model do not display this problem. In the present mechanism, the thermodynamic data for HCOOH were adopted from the work of Goldsmith et al. [40]. The thermodynamic data of HOCO radicals in the present model and in the MG model are the same from the Burcat database [39], while in the GK model they were taken from the earlier work of Fabian and Janoschek [41]. One may conclude that the deteriorated performance of the GK model compared to the MG model for formic acid flames is the result of the modifications in the thermodynamic data, and not in the rate constants of the key reactions which are the same in both mechanisms.

The SK model increasingly deviates from the present experimental data with an increased amount of formic acid in the fuel mixtures as seen in Figs. 1–3. To reveal reactions responsible for this, a sensitivity analysis has been performed for lean, stoichiometric, and rich flames depicted in Fig. 1 using the SK model. The sensitivity spectra shown in Fig. 5 include reactions important for all hydrocarbon flames, such as



as well as reactions pertinent to methane flames, e.g.,



For formic acid, the sensitivity analysis pointed out several key reactions controlling flame propagation, essentially confirming the

assessment of Marshall and Glarborg [19]. These reactions were extensively examined in the recent studies [24–26] and thus only briefly discussed here. Hydrogen abstraction from formic acid produces radicals HOCO and HCOO, and the overall consumption rate depends on the individual competition of their formation as seen in Fig. 5 for reactions



These reactions have relatively small sensitivity coefficients, yet with the opposite sign. The reason for that is the different fate of the primary radicals: HCOO radicals are short-living species, they decompose rapidly forming H atoms and none of the possible reactions with other radicals or with O₂ can compete with this decomposition. Thus, any reaction of abstraction from formic acid producing HCOO effectively yields H atoms. This is confirmed by the fact that the choice of the estimated rate constants of reactions



which are different in the MG model and in the SK model, does not affect the calculated burning velocity of formic acid.

HOCO radicals are much more stable, thus they can participate in the chain termination



or be converted into less reactive radicals in



Decomposition of these radicals also competitively occurs via two channels



from which the second one has the second largest sensitivity coefficients in all formic acid flames after the chain-branching reaction (4). It should be emphasised that the rate constants of all reactions between HCOOH and radicals adopted in the SK model and the present mechanism are the same as in the MG model [19]. The motivation of this choice has been given by Savchenkova et al. [29], and, to avoid redundancy, it is included in the Supplemental Material.

Further comparison of the SK model and MG model corroborates that the different behaviour in the modeling of formic acid flames is mostly due to different sub-mechanisms for HOCO radicals. Relevant reactions have been discussed by Nilsson and Konnov [42], where the role of HOCO chemistry in syngas combustion was analysed. It was noticed that many pertinent rate constants in the models developed in Lund are the same as in the MG model, however, for the key reaction of the HOCO decomposition they were adopted from different sources. The rate constant of reaction (15) selected in [42] was taken from the study of Larson et al. [43], who analysed reactions (14) and (15) together with the reaction between CO and OH using available experimental data and theoretical two-channel RRKM model. A decade later Golden et al. [38] revisited this system and performed dedicated measurements of the rate constants of CO + OH at low (253–343 K) and high (1400–2600 K) temperatures. Furthermore, a two-channel RRKM model

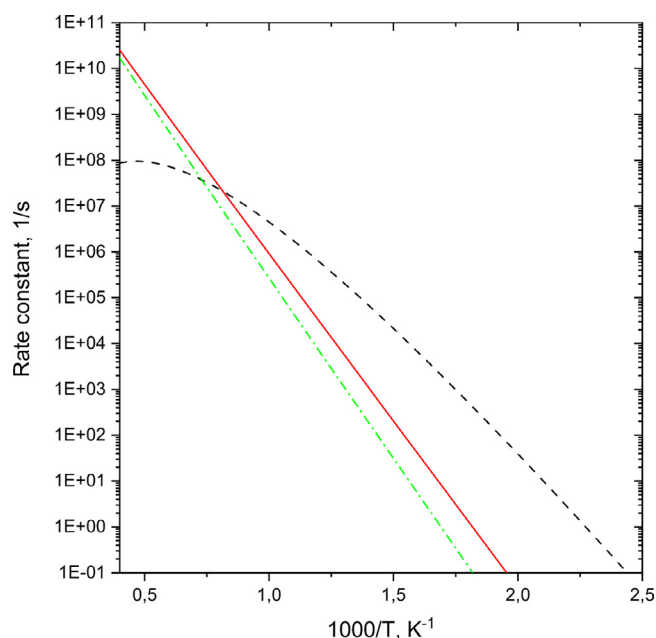


Fig. 6. High-pressure rate constants of reaction $\text{HOCO}(+M)=\text{H}+\text{CO}_2(+M)$. Red line – Larson et al. [43], green line – Golden et al. [38], black line – rate constant at 987 atm from Aramco 2.0. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

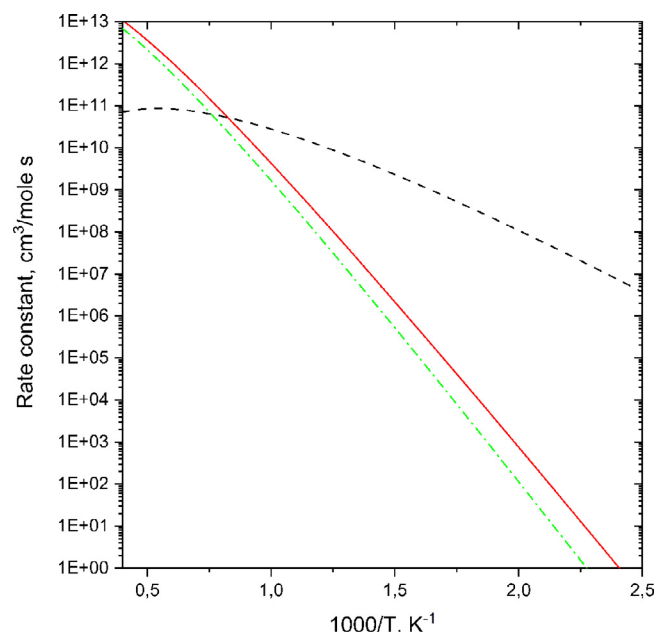


Fig. 7. Low-pressure rate constants of reaction $\text{HOCO}(+M)=\text{H}+\text{CO}_2(+M)$. Red line – Larson et al. [43], green line – Golden et al. [38], black line – rate constant at 0.001 atm from Aramco 2.0. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

was constructed and optimised using previous and new experimental data. The rate constant of reaction (15) derived by Golden et al. [38] was found significantly lower than that of Larson et al. [43]. The difference is about a factor of 2 at 2000 K increasing to almost an order of magnitude at 500 K for both high-pressure and low-pressure limit rate constants as illustrated in Figs. 6 and 7, respectively.

The multichannel reaction $\text{CO}+\text{OH}$ was a subject of numerous studies summarised in 2008 by Smith [44] and later scrutinized using ab initio theoretical works, e.g., [45,46], and experiments,

e.g., [47,48], however with the focus on carbon monoxide oxidation by OH without providing new expressions for the rate constant of HOCO decomposition. Nilsson and Konnov [42] in 2015 noticed that many contemporary models do not include HOCO. AramcoMech 2.0 and AramcoMech 3.0 implemented the rate constant of reaction (15) in PLOG format, the source of which is not documented. It should be mentioned that recent models developed at Lawrence Livermore National Laboratory and Politecnico di Milano include AramcoMech as a core C0 – C3 mechanism. The PLOG terms for reaction (15) from AramcoMech at 0.001 atm and 987 atm dramatically deviate from the low-pressure and high-pressure limit rate constants [38,43], respectively, as also illustrated in Figs. 6 and 7. This deviation was a major reason for significant modification of the calculated burning velocity when pertinent reactions (called Part III by Yin et al. [25]) were replaced in AramcoMech 3.0 by the reactions taken from the MG model.

A straightforward replacement of the previous rate constant from [43] in the present mechanism by the rate expression of reaction (15) derived by Golden et al. [38] dramatically reduces calculated burning velocities of $\text{HCOOH} + \text{CH}_4 + \text{air}$ flames as shown in Fig. 1 and yields significant underprediction as compared to the present measurements and other burning velocities of formic acid flames from the literature. A better agreement of the model with the experimental data could be achieved by modifying the rate constants of several reactions pertinent to formic acid and HOCO chemistry within their uncertainties evaluated earlier [29,42]. However, due to the relatively low sensitivity of the burning velocities to these parameters as seen in Fig. 5, this would require modifications to the limits of reasonable uncertainty ranges and was not attempted in the present work. The exception is reaction (15) whose rate constant from Golden et al. [38] in the present work was increased by a factor of two, well within the divergence with the expression of Larson et al. [43] discussed above. It should be noted that the optimized parameters derived from the two-channel RRKM calculations were presented for N_2 as a bath gas, and at atmospheric and higher pressures the rate constants are in the falloff region [38]. Thus, the effective increase of the rate constant of reaction (15) could be achieved in the model introducing different collisional efficiencies, e.g., for H_2O , Ar, and other main components. However, notable data scattering in the measurements of $\text{HCOOH} + \text{O}_2 + \text{N}_2$ flames and significant experimental uncertainties of the burning velocities of $\text{HCOOH} + \text{O}_2 + \text{Ar}$ flames [26] discussed in the following prevent from conclusive parametrization of this rate constant using collisional efficiencies, which are thus all equal to unity in the present model.

The suggested modification of the rate constant of reaction (15) yields a very good agreement with the present experimental data as seen in Figs. 1, and 2 over entire ranges of equivalence ratio and of the mole fraction of formic acid in the fuel mixture for stoichiometric conditions, Fig. 3. For $\text{HCOOH} + (0.4 \text{ O}_2 + 0.6 \text{ N}_2)$ flames at 453 K shown in Fig. 4, the present mechanism demonstrates close performance to the MG model in lean mixtures, while in rich mixtures these two models diverge. Similar behaviour is seen for $\text{HCOOH} + (0.35 \text{ O}_2 + 0.65 \text{ N}_2)$ flames at 453 K presented in Fig. 8. In both cases, the present model is in better agreement with the experimental data of Yin et al. [25] from moderately lean to moderately rich flames compared to the MG model.

The spread of the datapoints in very rich mixtures seen in Figs. 4 and 8 reaches 5–8 cm/s and could be attributed to flame instabilities, non-homogeneous mixture preparation, and other experimental difficulties associated with the investigation of liquid fuels in spherical flames [25]. It is therefore interesting to compare model behaviour with the measurements performed using a different method. Figure 9 presents the burning velocities of $\text{HCOOH} + \text{O}_2 + \text{Ar}$ flames at 368 K obtained in conical flames [26]. As is mentioned above, the argon fraction was fixed at 0.55, while

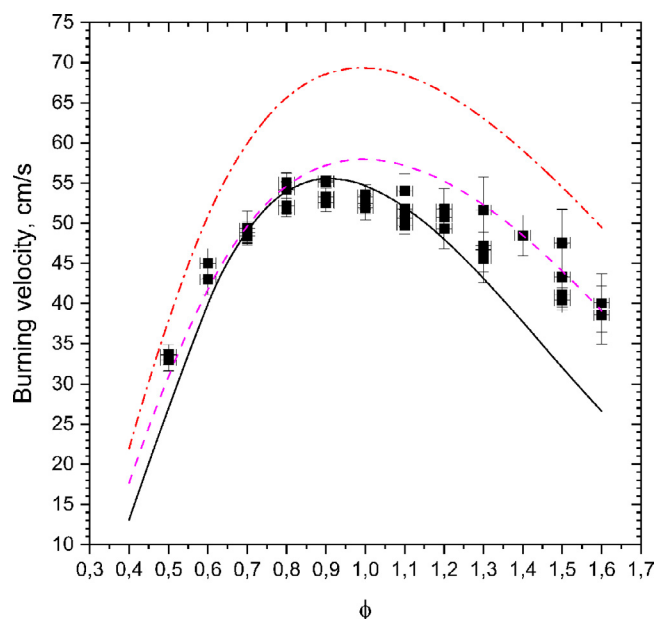


Fig. 8. Laminar burning velocities of $\text{HCOOH} + (0.35 \text{ O}_2 + 0.65 \text{ N}_2)$ flames at 453 K. Symbols: experiments [25]; dash-dot red line – GK model, dashed magenta line – MG model, solid black line – present model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

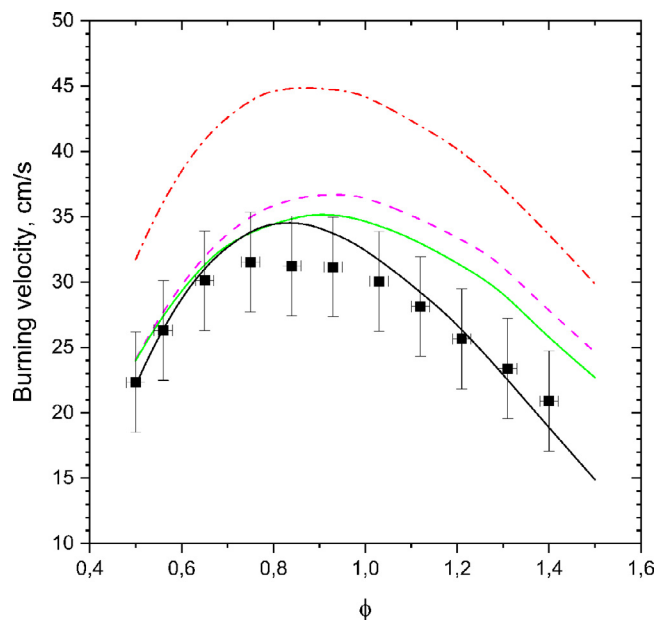


Fig. 9. Laminar burning velocities of $\text{HCOOH} + \text{O}_2 + \text{Ar}$ flames at 368 K. Mole fraction of Ar is 0.55. Symbols: experiments [26]; dash-dot red line – GK model, dashed magenta line – MG model, solid green line – merged model (AramcoMech 2.0 + MG model) [24], solid black line – present model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

amounts of formic acid and oxygen varied according to the equivalence ratio. Osipova et al. [26] found that the model merged by Sarathy et al. [24] from the AramcoMech 2.0 and the formic acid reaction subset from [19] predicts slightly lower burning velocities than the MG model as seen in Fig. 9. The present model agrees well with the MG model in lean flames and yields lower burning velocities in rich flames similar to the performance observed in Figs. 4 and 8. This brings an excellent agreement with the measurements [26] over the entire range of equivalence ratios.

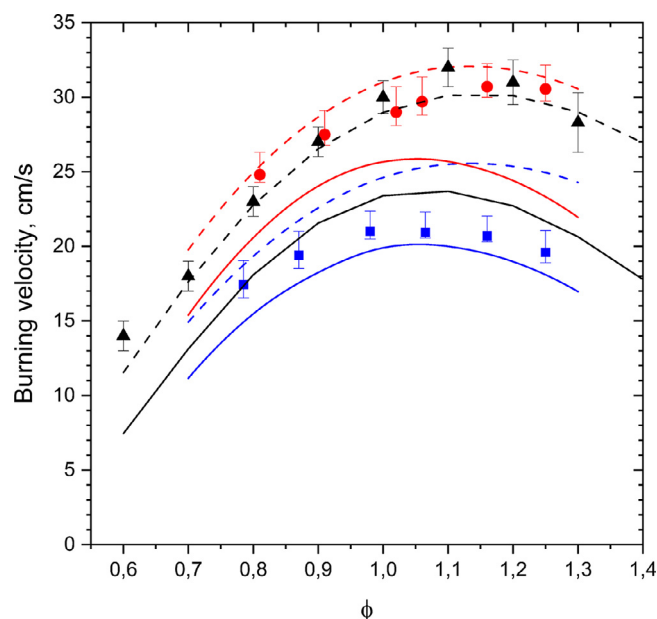


Fig. 10. Laminar burning velocities of different blends at 1 atm. Symbols: experiments [24], dashed lines – MG model, solid lines – present model. Triangles: $(0.1 \text{ H}_2 + \text{HCOOH}) + \text{air}$ at 358 K, circles and squares: $\text{HCOOH} + \text{air}$ flames at 373 and 423 K, respectively.

The comparison of the present model and the MG model with the experimental data from Sarathy et al. [24] shown in Fig. 10 looks confusing. One mechanism is in a better agreement with the burning velocities of formic acid + air mixtures at 373 K, while another – with those at 423 K. The measurements of $(0.1 \text{ H}_2 + 0.9 \text{ HCOOH}) + \text{air}$ flames at 358 K are also closely reproduced by the MG model. One should note that these measurements have been performed in different laboratories, namely at 373 and 423 K – at PRISME, and at 358 K – at KAUST, thus the data inconsistency cannot be simply explained by the differences in the experimental equipment or stretch correction methods used. In principle, an improved agreement for all datasets shown in Fig. 10 could be achieved by reducing the overall reactivity of the MG model, for example by merging the formic acid reactions from the MG model with the AramcoMech 2.0 as was suggested by Sarathy et al. [24]. Nevertheless, the calculated burning velocities predicted by this compiled model (presented in [24]) do not overlap with the measurements within the error bars, which indicates that evaluated experimental uncertainties are too optimistic for the challenging experiments with formic acid flames. Analysis of the temperature dependence of the results obtained at 373 and 423 K further confirms the problems with the data consistency. Simple empirical correlation $S_L/S_{L0} = (T/T_0)^\alpha$ relates the burning velocity, S_L , at the initial temperature T , with the burning velocity, S_{L0} , at a reference temperature T_0 . Power exponents, α , moderately depend on the equivalence ratio, ϕ , usually with a minimum in the mixtures possessing maximum burning velocity [49]. An important feature of this temperature dependence analysis is that the power exponents predicted by different models for the same fuel are very close to each other even when the calculated burning velocities diverge, as was first found by Christensen et al. [50] and later demonstrated for many fuels [49]. Indeed, for $\text{HCOOH} + \text{air}$ flames at 373 and 423 K, the MG model predicts the ratio $S_L/S_{L0} = 1.26$ at $\phi = 1$ and 1255 at $\phi = 1.2$, while the present model predicts the ratio $S_L/S_{L0} = 1.28$ at $\phi = 1$ and 1286 at $\phi = 1.2$. On the contrary, the experimental ratio $S_L/S_{L0} = 1.38$ at ϕ close to 1 and 156 at $\phi = 1.25$, thus deviating by more than 20% from the predictions of two models.

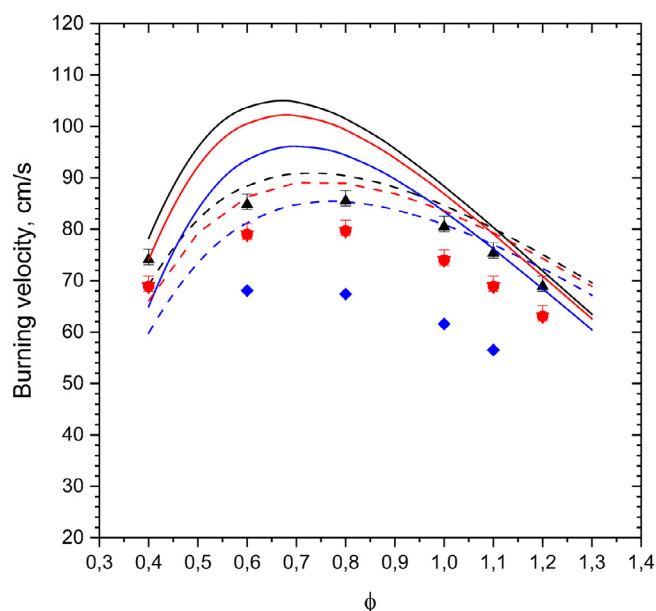


Fig. 11. Laminar burning velocities of $\text{HCOOH} + \text{O}_2 + \text{N}_2$ flames at 433 K. Symbols: experiments [20]; dashed lines – MG model, solid lines – present model. Triangles: 12% N_2 , circles: 16% N_2 , squares: 24% N_2 in the oxidizer.

Finally, it is instructive to compare the first measurements of the formic acid burning velocity by De Wilde and Van Tiggelen [20] performed at 433 K with predictions of the MG model and the present model. To obtain stable conical flames the amount of oxygen in ($\text{O}_2 + \text{N}_2$) oxidizer mixtures was varied from 76 to 88%; three series of these experiments are presented in Fig. 11. Marshall and Glarborg [19] noticed that their model agrees satisfactorily with lower diluted flames (12% N_2), while for 24% N_2 mixtures overprediction reaches 10–15 cm/s. The present model overpredicts the burning velocities for all mixtures. Remarkably, all models, including the merged model (AramcoMech 2.0 + MG model) [24], predict a very small variation of the burning velocity with dilution, which is close to 6–7 cm/s between the mixtures with 12 and 24% of N_2 at equivalence ratio 0.8, while experimentally this difference was found to be about 18 cm/s [20]. This is a strong indication of possible experimental problems not fully accounted for. The temperature measurements performed in these flames [20] also show significant variation from 2336 K in the mixture containing 12% N_2 down to 2252 K for 24% of N_2 at $\phi = 0.8$, while the calculated adiabatic flame temperatures reduce from 2428 K down to 2400 K. One can speculate that nitrogen flow used to introduce Na for the temperature measurements using a reversal of the sodium doublet method may contain non-negligible amounts of steam thus affecting the flame temperature and the burning velocity, and making this dataset unsuitable for the model validation.

5. Conclusions

Recent measurements of the burning velocity of formic acid mixtures in spherical flames revealed significant experimental difficulties which, even with all precautions taken, lead to significant scatter of the data in rich flames [25], and abnormal variation with the mixture temperature [24]. Similar problems were reported for propionic acid flames [23], where day-to-day disparities indicate interaction with the surface of combustion bombs. Therefore, new measurements of the laminar burning velocities of methane + formic acid + air flames using the heat flux method at 1 atm are advantageous since, in the flow regime, the sur-

face effects are reduced due to short time of contact and adsorption/desorption processes deemed to be balanced. A comparison of the new and data from the literature with predictions of kinetic models revealed that the mechanism of Glarborg et al. [34] and the previous model of the authors [29] significantly overpredict the burning velocities of formic acid and its mixtures. A modification of the rate constant of reaction $\text{HOCO}(\text{+M})=\text{H}+\text{CO}_2(\text{+M})$ was suggested that dramatically improved the present model performance. For lean flames of formic acid, this model shows good agreement with the model developed by Marshall and Glarborg [19], while in rich flames significant scatter of the measurements prevents from a conclusive endorsement of one or another mechanism. The present analysis calls for further investigation of the rate constants of HOCO radicals and independent experimental studies of HCOOH flames.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors acknowledge funding from the Knut and Alice Wallenberg Foundation through grant KAW2019.0084 COCALD.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2020.10.050.

References

- [1] P.C. Chebbi, Carboxylic acids in the troposphere, occurrence, sources, and sinks: a review, *Atmos. Environ.* 30 (1996) 4233–4249.
- [2] R.W. Talbot, B.W. Mosher, B.G. Heikes, D.J. Jacob, J.W. Munger, B.C. Daube, W.C. Keene, J.R. Maben, R.S. Artz, Carboxylic acids in the rural continental atmosphere over the eastern United States during the Shenandoah cloud and photochemistry experiment, *J. Geophys. Res. Atmos.* 100 (D5) (1995) 9335–9343.
- [3] K. Kawamura, L.L. Ng, I.R. Kaplan, Determination of organic acids ($\text{C}_1 - \text{C}_{10}$) in the atmosphere, motor exhausts, and engine oils, *Environ. Sci. Technol.* 19 (1985) 1082–1086.
- [4] J.E. Lawrence, P. Koutrakis, Measurement of atmospheric formic and acetic acids: methods evaluation and results from field studies, *Environ. Sci. Technol.* 28 (5) (1994) 957–964.
- [5] R. Castedo, R.M. Peña, C. Herrero, S. Garci-Martín, Analysis of formic and acetic acid in rain water by capillary electrophoresis, *Int. J. Environ. Anal. Chem.* 83 (3) (2003) 247–253.
- [6] E. Zervas, X. Montagne, J. Lahaye, Emission of specific pollutants from a compression ignition engine. Influence of fuel hydro-treatment and fuel/air equivalence ratio, *Atmos. Environ.* 35 (2001) 1301–1306.
- [7] E. Zervas, X. Montagne, J. Lahaye, $\text{C}_1 - \text{C}_5$ organic acid emissions from an SI engine: influence of fuel and air/fuel ratio, *Environ. Sci. Technol.* 35 (2001) 2746–2751.
- [8] E. Zervas, M. Tazerout, Organic acids emissions from natural-gas-fed engines, *Atmos. Environ.* 34 (23) (2000) 3921–3929.
- [9] E. Zervas, Formation of organic acids from propane, isooctane, and toluene/isooctane flames, *Fuel* 84 (2005) 691–700.
- [10] E. Zervas, S. Pouloupoulos, C. Philippopoulos, Formation of oxygenated compounds from isooctane flames, *Fuel* 85 (3) (2006) 333–339.
- [11] F. Battin-Leclerc, A.A. Konnov, J.L. Jaffrezou, M. Legrand, To better understand the formation of short-chain acids in combustion systems, *Combust. Sci. Technol.* 180 (2008) 343–370.
- [12] X. Zhang, G. Wang, J. Zou, Y. Li, W. Li, T. Li, H. Jin, Z. Zhou, Y.Y. Lee, Investigation on the oxidation chemistry of methanol in laminar premixed flames, *Combust. Flame* 180 (2017) 20–31.
- [13] Demirbas, The influence of temperature on the yields of compounds existing in bio-oils obtained from biomass samples via pyrolysis, *Fuel Process. Technol.* 88 (6) (2007) 591–597.
- [14] F. Valentini, V. Kozell, C. Petrucci, A. Marroccoli, Y. Gu, D. Gelman, L. Vaccaro, Formic acid, a biomass-derived source of energy and hydrogen for biomass upgrading, *Energy Environ. Sci.* 12 (9) (2019) 2646–2664.

- [15] K. Müller, K. Brooks, T. Autrey, Hydrogen storage in formic acid: a comparison of process options, *Energy Fuels* 31 (11) (2017) 12603–12611.
- [16] J. Eppinger, K.-W. Huang, Formic acid as a hydrogen energy carrier, *ACS Energy Lett.* 2 (2017) 188–195.
- [17] X. Wang, Q. Meng, L. Gao, Z. Jin, J. Ge, C. Liu, W. Xing, Recent progress in hydrogen production from formic acid decomposition, *Int. J. Hydrogen Energy* 43 (14) (2018) 7055–7071.
- [18] K. Sordakis, C. Tang, L.K. Vogt, H. Junge, P.J. Dyson, M. Beller, G. Laurenczy, Homogeneous catalysis for sustainable hydrogen storage in formic acid and alcohols, *Chem. Rev.* 118 (2018) 372–433.
- [19] P. Marshall, P. Glarborg, Ab initio and kinetic modeling studies of formic acid oxidation, *Proc. Combust. Inst.* 35 (2015) 153–160.
- [20] E. De Wilde, A. Van Tiggelen, Burning velocities in mixtures of methyl alcohol, formaldehyde or formic acid with oxygen, *Bull. Soc. Chim. Belges* 77 (1968) 67–76.
- [21] N. Leplat, J. Vandooren, Numerical and experimental study of the combustion of acetic acid in laminar premixed flames, *Combust. Flame* 159 (2012) 493–499.
- [22] M. Christensen, A.A. Konnov, Laminar burning velocity of acetic acid + air flames, *Combust. Flame* 170 (2016) 12–29.
- [23] X. Zhang, B. Mei, S. Ma, Y. Zhang, C. Cao, W. Li, L. Ye, Y. Li, Characterizing the fuel-specific combustion chemistry of acetic acid and propanoic acid: laminar flame propagation and kinetic modeling studies, *Proc. Combust. Inst.* (2020), doi:10.1016/j.proci.2020.06.220.
- [24] S.M. Sarathy, P. Brequigny, A. Katoch, A.M. Elbaz, W.L. Roberts, R.W. Dibble, F. Foucher, Laminar burning velocities and kinetic modeling of a renewable e-fuel: formic acid and its mixtures with H₂ and CO₂, *Energy Fuels* 34 (2020) 7564–7572.
- [25] G. Yin, Q. Gao, E. Hu, J. Xu, M. Zhou, Z. Huang, Experimental and kinetic study on laminar flame speeds of formic acid, *Combust. Flame* 220 (2020) 73–81.
- [26] K.N. Osipova, S.M. Sarathy, O.P. Korobeinichev, A.G. Shmakov, Chemical structure of atmospheric pressure premixed laminar formic acid/hydrogen flames, *Proc. Combust. Inst.* (2021) doi:10.1016/j.proci.2020.06.033.
- [27] A.S. Semnikhin, E.G. Shubina, A.S. Savchenkova, I.V. Chechet, S.G. Matveev, A.A. Konnov, A.M. Mebel, Mechanism and rate constants of the CH₃ + CH₂CO reaction: a theoretical study, *Int. J. Chem. Kinet.* 50 (2018) 273–284.
- [28] A.S. Savchenkova, A.S. Semnikhin, I.V. Chechet, S.G. Matveev, A.A. Konnov, A.M. Mebel, Mechanism and rate constants of the CH₂ + CH₂CO reactions in triplet and singlet states: a theoretical study, *J. Comput. Chem.* 40 (2019) 387–399.
- [29] A.S. Savchenkova, A.S. Semnikhin, I.V. Chechet, S.G. Matveev, A.M. Mebel, A.A. Konnov, Revisiting diacetyl and acetic acid flames: the role of the ketene + OH reaction, *Combust. Flame* 218 (2020) 28–41.
- [30] L.P.H. de Goey, A. van Maaren, R.M. Quax, Stabilization of adiabatic premixed laminar flames on a flat flame burner, *Combust. Sci. Technol.* 92 (1993) 201–207.
- [31] V.A. Alekseev, J.D. Nacler, M. Christensen, E.J.K. Nilsson, E.N. Volkov, L.P.H. de Goey, A.A. Konnov, Experimental uncertainties of the heat flux method for measuring burning velocities, *Combust. Sci. Technol.* 188 (2016) 853–894.
- [32] R.T.E. Hermanns, Laminar Burning Velocities of Methane-Hydrogen-Air Mixtures Ph.D. thesis, Eindhoven University of Technology, Eindhoven, the Netherlands, 2007.
- [33] ANSYS CHEMKIN 17.0 (15151) 2016, ANSYS reaction design, San Diego, USA.
- [34] P. Glarborg, J.A. Miller, B. Ruscic, S.J. Klippenstein, Modeling nitrogen chemistry in combustion, *Prog. Energy Combust. Sci.* 67 (2018) 31–68.
- [35] G. Capriolo, A.A. Konnov, Combustion of propanol isomers: experimental and kinetic modelling study, *Combust. Flame* 218 (2020) 189–204.
- [36] G. Capriolo, C. Brackmann, M. Lubrano Lavadera, T. Methling, A.A. Konnov, An experimental and kinetic modelling study on nitric oxide formation in premixed C3 alcohols flames, *Proc. Combust. Inst.* (2021), doi:10.1016/j.proci.2020.07.051.
- [37] X. Han, M. Lubrano Lavadera, C. Brackmann, Z. Wang, Y. He, A.A. Konnov, Experimental and kinetic modelling study of NO formation in premixed CH₄+O₂+N₂ flames, *Combust. Flame* 223 (2021) 349–360.
- [38] D.M. Golden, G.P. Smith, A.B. McEwen, C.-L. Yu, B. Eiteneer, M. Frenklach, G.L. Vaghjani, A.R. Ravishankara, F.P. Tully, OH (OD)+ CO: measurements and an optimized RRKM fit, *J. Phys. Chem. A* 102 (1998) 8598–8606.
- [39] E. Goos, A. Burcat, B. Ruscic, Ideal gas thermochemical database with updates from active thermochemical. <http://burcat.technion.ac.il/dir/> Accessed April 2016, 2020
- [40] C.F. Goldsmith, G.R. Magoon, W.H. Green, Database of small molecule thermochemistry for combustion, *J. Phys. Chem. A* 116 (36) (2012) 9033–9057.
- [41] W.M. Fabian, R. Janoschek, Thermochemical properties of the hydroxy-formyl radical, HO–CO, and the formyloxy radical, HC (O) O, and their role in the reaction OH+ CO→ H+ CO₂. Computational G3MP2B3 and CCSD (T)-CBS studies, *J. Mol. Struct. THEOCHEM* 713 (2005) 227–234.
- [42] E.J.K. Nilsson, A.A. Konnov, Role of HOCO chemistry in syngas combustion, *Energy Fuels* 30 (2016) 2443–2457.
- [43] C.W. Larson, P.H. Stewart, D.M. Golden, Pressure and temperature dependence of reactions proceeding via a bound complex. An approach for combustion and atmospheric chemistry modelers. Application to HO+ CO→[HOCO]→ H+ CO₂, *Int. J. Chem. Kinet.* 20 (1) (1988) 27–40.
- [44] I.W. Smith, The temperature-dependence of elementary reaction rates: beyond Arrhenius, *Chem. Soc. Rev.* 37 (4) (2008) 812–826.
- [45] T.L. Nguyen, B.C. Xue, R.E. Weston Jr, J.R. Barker, J.F. Stanton, Reaction of HO with CO: tunneling is indeed important, *J. Phys. Chem. Lett.* 3 (11) (2012) 1549–1553.
- [46] R.E. Weston Jr, T.L. Nguyen, J.F. Stanton, J.R. Barker, HO+ CO reaction rates and H/D kinetic isotope effects: master equation models with ab initio SCTST rate constants, *J. Phys. Chem. A* 117 (5) (2013) 821–835.
- [47] Y. Liu, S.P. Sander, Rate constant for the OH+ CO reaction at low temperatures, *J. Phys. Chem. A* 119 (39) (2015) 10060–10066.
- [48] E.F. Nasir, A. Farooq, A shock-tube study of the CO+ OH reaction near the low-pressure limit, *J. Phys. Chem. A* 120 (22) (2016) 3924–3928.
- [49] A.A. Konnov, M. Akram, V. Ratna Kishore, N.I. Kim, C. Prathap, S. Kumar, A comprehensive review of measurements and data analysis of laminar burning velocities for various fuel+air mixtures, *Prog. Energy Combust. Sci.* 68 (2018) 197–267.
- [50] M. Christensen, E.J.K. Nilsson, A.A. Konnov, The temperature dependence of the laminar burning velocities of methyl formate + air flames, *Fuel* 157 (2015) 162–170.